

Test Yourself
Microeconomics
Basic Economics +
Production Possibility
Curve

Section A

Duration of each question is **40 mins**

1. With the help of a diagram, explain the difference between the causes of a movement along, and a shift of, a production possibility curve (PPC) **and** consider which is likely to have the most immediate impact on an economy. (8) **q2a/21/M/J/23**
2. Entrepreneurs are vital in an economy to achieve long run economic growth. Assess whether encouraging entrepreneurship or increasing another of the factors of production will be a more effective way to achieve long-run economic growth. (12) **q3b/22/M/J/23**
3. Assess whether producers are the only ones to benefit when an economy decides to allocate additional resources to investment. (12) **q3b/23/M/J/24**
4. With the help of a diagram, explain the significance of a position within a market economy's production possibility curve (PPC) **and** consider whether such a position is likely to be permanent. (8) **q3a/21/O/N/24**
5. With the help of a production possibility curve (PPC) diagram(s), explain the difference between constant and increasing opportunity costs **and** consider how choices in deciding which type of goods to produce in the short run may influence future economic growth. (8) **q3a/22/O/N/24**
6. With the help of a production possibility curve (PPC) diagram, explain the terms scarcity and choice **and** consider the extent to which every choice has an equal opportunity cost. (8) **q2a/22/F/M/25**
7. An economy contains just two sectors, an agricultural sector and an industrial sector. With the help of a diagram, explain how a production possibility curve (PPC) can be used to show the concept of opportunity cost **and** consider whether opportunity cost is likely to remain constant if resources are reallocated from the agricultural sector to the industrial sector. (8) **q2a/specimen23**

Section B

Duration 1 hour

1. A worker earns \$40 per hour. Rather than work, she decides to visit a museum for three hours. The visit costs a total of \$40.

What is the opportunity cost of visiting the museum?

- A** \$40 **B** \$80 **C** \$120 **D** \$160

2. The nature of a typical car assembly plant has changed. The industry has fewer firms, operates on larger sites and has more automated machinery.

How is this change most likely to have affected the relative use of factors of production in the industry?

	increased relative use	decreased relative use
A	capital and enterprise	labour and land
B	enterprise and labour	land and capital
C	labour and land	capital and enterprise
D	land and capital	enterprise and labour

3. Why does the concept of scarcity apply to the use of fossil fuels?

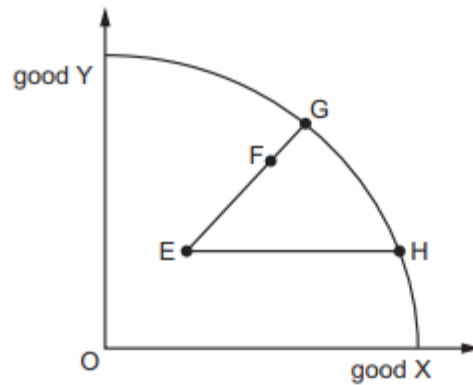
- A** Demand fluctuates according to price changes.
B Supply is insufficient to meet demand.
C Their use is restricted because of harmful pollution.
D They are being replaced by renewable energy sources.

4. An economist knows the current point at which an economy operates within its production possibility curve.

What can the economist conclude about this economy?

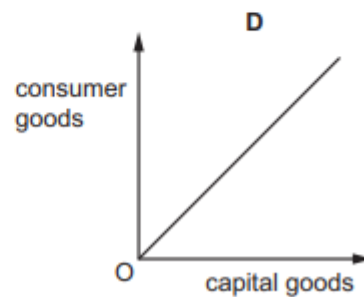
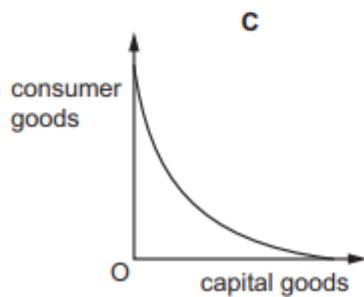
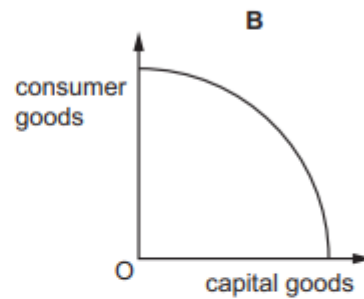
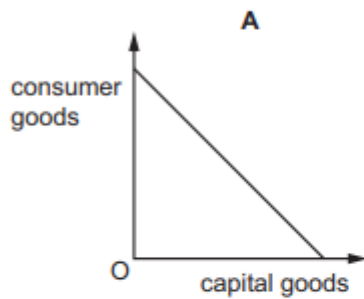
- A** its degree of self-sufficiency
B its international competitiveness
C its level of output of two goods
D its rate of economic growth

5. The diagram shows a production possibility curve for an economy that produces two goods, X and Y.



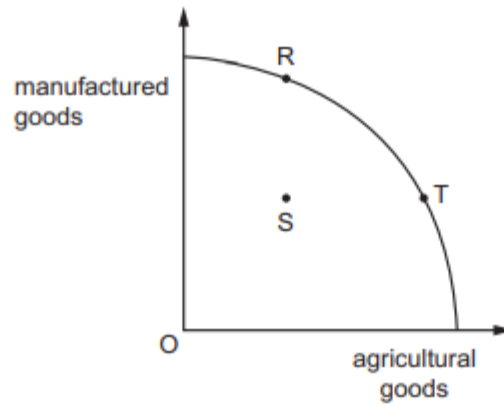
When will the opportunity cost of producing more of good X be the largest?

- A moving from point E to point F
 - B moving from point E to point G
 - C moving from point E to point H
 - D moving from point G to point H
6. The diagrams show production possibility curves.
- Which diagram shows constant opportunity costs?

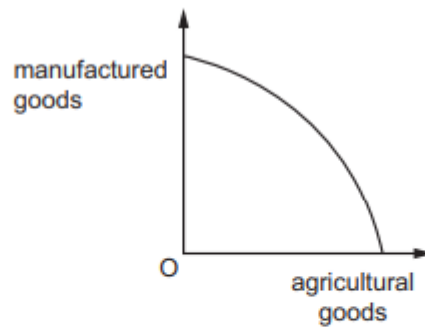


7. Farmers using traditional methods lack access to finance and often employ family members on a part-time basis.

If working practices in agriculture could be improved, how would this be likely to be shown on the production possibility curve?



- A by a movement from R to T
 - B by a movement from S to R
 - C by a movement from S to T
 - D by a movement from T to R
8. The production possibility curve for a country is shown.



What can be determined from the diagram?

- A the consumers' preferred combination of output
- B the level of economic growth
- C the opportunity cost of manufactured goods in terms of agricultural products
- D the preference for present consumption rather than future consumption

9. The factors of production earn different rewards.

What identifies the correct economic term for these rewards?

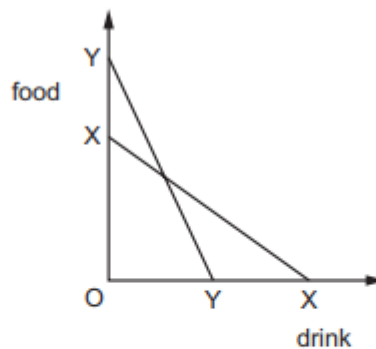
- A capital – surpluses
- B enterprise – dividends
- C labour – interest
- D land – rents

10. A hospital management board decides to reallocate \$800 000 from its \$1.2m new buildings budget to spend on cancer care.

What is the opportunity cost of this decision?

- A a cost of \$400 000
- B a cost of \$800 000
- C the cost of healthcare for cancer patients
- D the loss of some new buildings

11. The diagram shows the change in a country's production possibility curve from XX to YY.

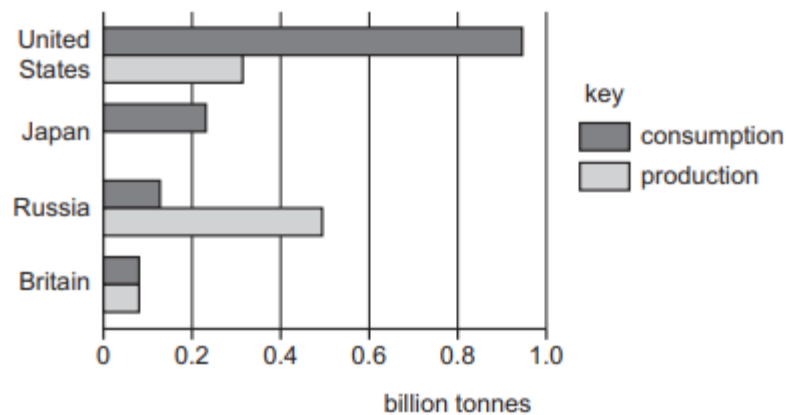


What would explain this change?

- A Consumers chose to consume more food and less drink.
- B Government taxed food production and subsidised drink production.
- C Productivity rose in food production and fell in drink production.
- D There were more imports of food and more exports of drink.

12. Assuming nothing else changes, which change in an economy's labour market will cause the production possibility curve to shift to the left?
- A an increase in worker immigration
 - B an increase in the retirement age
 - C an increase in labour productivity
 - D an increase in the school leaving age

13. There is a worldwide shortage of oil. The diagram shows the consumption and production of oil (billion tonnes) for selected countries.



What can be concluded from the diagram?

- A In Britain, oil has zero opportunity cost.
 - B In Japan, there is self-sufficiency in oil.
 - C In Russia, there is no shortage of oil.
 - D In the United States, rationing is the only solution to oil shortage.
14. A government decides that all non-essential businesses must close for three months due to a disease pandemic.

What is an opportunity cost of this policy?

- A increased leisure time for employees
- B lives that are saved due to reduced disease levels
- C the production of goods by essential businesses
- D the loss of goods from the closed businesses

15. The Indian government operates a pension scheme.

Which statement about this scheme would be classed as normative?

- A To be eligible, individuals must be above the age of 60 and live below the poverty line.
- B In January 2020, eligible individuals aged over 70 received 500 rupees per month.
- C In 2018, the Supreme Court said the government must review pension payments because they are unrealistic.
- D In January 2020, the Indian government announced that it might increase the payment to 1000 rupees per month.

16. Economics students have to fill in the gaps in the following passage.

'The three basic questions arising from the economic problem of scarcity are1..... to produce,2..... to produce, and3..... to produce.'

Which combination is correct?

	1	2	3
A	how	when	where
B	how	where	for whom
C	what	how	for whom
D	what	when	where

17. Which list has the most words that are more likely to relate to a positive statement than to a normative statement?

- A describe, better, should, value, emotion
- B bias, objective, ought, opinion, tastes
- C contestable, is, motive, subjective, moral
- D actual, ethics, fact, was, testable

18. The table shows the amount of direct taxes that different households in the UK paid in 2017–2018.

	bottom 20% of households	second 20% of households	middle 20% of households	fourth 20% of households	top 20% of households
average income (£)	15 300	25 000	34 600	47 600	90 600
direct tax paid (£)	2 000	3 500	6 000	9 500	21 200

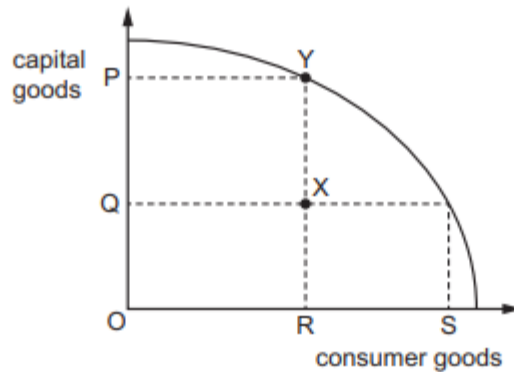
Which positive economic statement can be made from the table?

- A Direct tax is unfair on the richer households in the UK.
 - B Direct tax in the UK is a proportional tax.
 - C Poorer households do not pay enough direct tax in the UK.
 - D Rich households pay more direct tax than poor households in the UK.
19. Which term assumes the existence of 'qualities, knowledge and skills' that allow an individual to be more productive?
- A enterprise investment
 - B human capital
 - C mechanisation
 - D specialisation
20. A doctor has very long working hours and a high level of stress. She can become a teacher instead of continuing as a doctor.

What is the opportunity cost of choosing to continue as a doctor?

- A the risk of mistakes caused by tiredness
- B the possible costs of illness caused by high stress levels
- C the potential salary as a teacher
- D the retraining costs of becoming a teacher

21. The diagram shows an economy's production possibility curve (PPC). The economy has moved from X to Y.



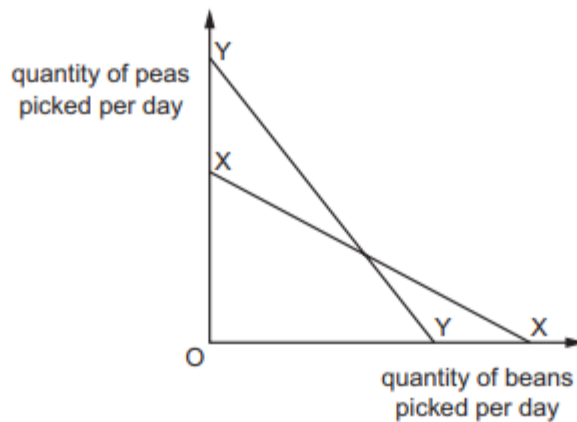
What can be deduced from the diagram?

- A Current living standards have fallen.
 - B The economy has become more efficient.
 - C There has been a rise in unemployment.
 - D There has been an opportunity cost of OR consumer goods.
22. Which action would **not** raise the quality of the particular factor of production?
- A capital – operating a machine continuously at full capacity
 - B enterprise – providing risk management advice
 - C land – increasing the use of fertilisers
 - D labour – making apprenticeships and training compulsory
23. In 2015, the Chinese economy grew at 6.9%, the Indian economy at 7.6% and the US economy at 2.4%.

Which comment contains **two** positive statements?

- A Chinese economic growth exceeded US economic growth by 4.5% in 2015 but was 0.7% less than that in India.
- B Chinese economic growth was more than double that in the US in 2015 but was disappointing compared to that in India.
- C Economic growth in India was better than that of the US.
- D Economic growth in the US was too low in 2015 and economic growth in India was too high.

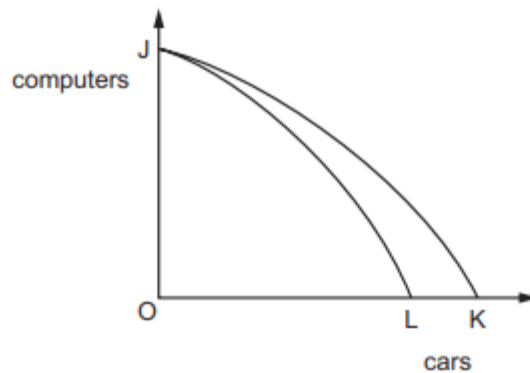
24. Line XX is the production possibility curve (PPC) of a worker picking peas and beans in a 10 hour working day.



What could cause the worker's PPC to shift to the line YY?

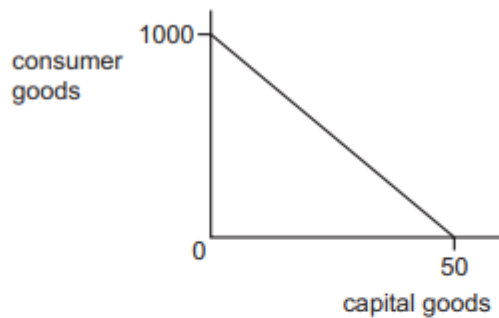
- A a decrease in working hours to 8 per day and a machine that increases the worker's pea picking productivity
 - B a machine that increases the worker's pea picking productivity only
 - C a new work schedule where the worker spends 6 hours per day picking peas and only 4 hours picking beans
 - D a reduction in working hours to 8 per day only
25. Which statement about the factors of production is correct?
- A The factor capital includes shares and deposit accounts at banks.
 - B The factor labour includes human-made resources.
 - C The factor land includes forests and undiscovered copper.
 - D The factor enterprise takes risks and is rewarded with interest.
26. Which statement is a positive statement?
- A Low inflation is preferable to low unemployment as a government aim.
 - B Low inflation will provide a more stable macro economy.
 - C The government must intervene in all markets to improve resource allocation.
 - D The government should cut income tax so living standards can improve.

27. In the diagram JK is the initial production possibility curve for an economy producing computers and cars.



What could cause the curve to shift to JL?

- A a fall in output per worker in the car industry
 - B a fall in the level of competition in the car industry
 - C a rise in the potential workforce of the economy
 - D a rise in the price of cars
28. The diagram shows an economy's production possibility curve.

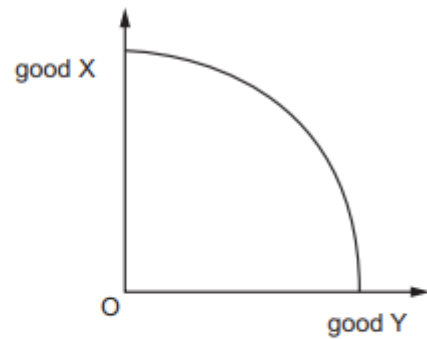


It has been employing its resources in the ratio of 80% consumer goods production and 20% capital goods production.

What will be the result if it decides to double its output of capital goods?

- A a gain of 20 capital goods
- B a gain of 40 capital goods
- C a loss of 200 consumer goods
- D a loss of 600 consumer goods

29. The diagram shows a production possibility curve.



What can be deduced from the shape of this curve?

- A decreasing marginal returns to consumption
 - B decreasing opportunity costs of consumption
 - C increasing opportunity costs of production
 - D increasing returns to scale
30. An economy is operating at a point inside its production possibility curve.
- Why is this described as inefficient?
- A Individuals are enjoying too much leisure.
 - B More of one good can be produced without decreasing production of another.
 - C The combination of labour and capital is wrong.
 - D There are shortages of some goods and an excess supply of others.

Question	Answer	Marks
1	With the help of a diagram, explain the difference between the causes of a movement along, and a shift of, a production possibility curve (PPC) <u>and</u> consider which is likely to have the most immediate impact on an economy. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and understanding (3 marks maximum) An understanding of a PPC shown through an accurate diagram, with correctly labelled axes (1), a movement along a PPC showing a trade-off between production on the two axes (1), and a shift of a PPC showing an increase in production of two goods with no trade-off. (1) AO2 Analysis (3 marks maximum) Uses the PPC to explain that a movement along a PPC is caused by a reallocation of resources (1), as resources are shifted between the two types of product, involving an opportunity cost (1), but a shift of a PPC is caused by an increase in the quantity and/or quality of resources, allowing more of both goods to be produced. (1) AO3 Evaluation (2 marks maximum) Offers a valid judgement on whether a movement along, or a shift of, a PPC is likely to have the most immediate impact on an economy (1) to reach a conclusion. (1)	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
2	Entrepreneurs are vital in a modern economy to achieve long run economic growth. Assess whether encouraging entrepreneurship or increasing another of the factors of production will be a more effective way to achieve long run economic growth. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis (max 8 marks) The advantages of encouraging entrepreneurship include: • The fact that entrepreneurs are the ones who organise the other three factors of production to produce output in an economy. • They are the ones who take risks and in return receive profit which is the motivation to be successful and so stimulate economic growth. • The fact that entrepreneurs must be creative and innovative to succeed and therefore should have the skills to anticipate current and future needs to be successful and therefore increase the productive potential of an economy and thus stimulate long run economic growth e.g., as measured by a shift to the right of the PPC. • Modern economies are dynamic, ever changing and the need to respond quickly to changes is one that can be met by successful entrepreneurs. BUT • There may be a cost in training and educating future entrepreneurs. • The rate of success of a policy to increase entrepreneurship may be low • The success depends on other factors such as the availability of government subsidies etc. which have an opportunity cost as well as the availability and quality of the other factors of production, • Because of high risks, entrepreneurs may seek short run gain which may not encourage long run economic growth and may in fact make it less likely e.g., due to rapid depletion of scarce resources. These should be compared with the likely importance and the advantage(s) and disadvantage(s) of increasing one other factor of production i.e., land OR labour OR capital. This may include an assessment of the availability of other factors of production. A one-sided response can only gain a maximum of Level 2.	12

Question	Answer	Marks
2	AO3 Evaluation (max 4 marks) - Consideration of the likely importance of increasing entrepreneurship compared to increasing one other factor of production to increase long run economic growth in a modern economy. - In conclusion, which factor of production is likely to be the most important. Accept all valid responses. A one-sided response cannot gain any marks for evaluation.	
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
3	Assess whether producers are the only ones to benefit when an economy decides to allocate additional resources to investment. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis - For an explanation / analysis of the benefits and costs to a producer e.g., producers will benefit as additional investment is likely to reduce costs of production, make them more price competitive both domestically and internationally, increase their profits etc. Costs may include the fact that taxes may have to rise to fund the additional investment, much of the growth will be international so businesses who focus on the domestic market may not benefit. - For an explanation/ analysis of the benefits and costs to the consumer e.g., in the long term this may lead to wider choice and lower prices but in the short term, taxes may increase to fund the additional investment, there may be fewer goods available to consumers and higher prices. - For an explanation/ analysis of the benefits and costs to a government e.g., an increase in tax revenue in the long-term but an increase in expenditure in the short-term. Level 1 responses will be assertive and lacking in explanations / mainly descriptive and/or or mainly lacking in relevance to the question. Level 2 responses may contain some inaccuracies and will be one sided. Analysis will be explained at least in part and will be largely relevant to the question. Level 3 responses will consider alternative policies / concepts etc. and will be balanced. Explanations of points raised will be offered and will be accurate and relevant to the question AO3 Evaluation For an assessment of whether producers are the only ones to benefit from additional investment compared to consumers and the government which may depend on the level of development of economy and extent of government involvement. Accept all valid responses. A one-sided response cannot gain any marks for evaluation. Please use a text box to show the mark split e.g., L2 5 L1 2	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
4	With the help of a diagram, explain the significance of a position within a market economy's production possibility curve (PPC) <u>and</u> consider whether such a position is likely to be permanent. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding, up to 3 marks for AO2 Analysis, and up to 2 marks for AO3 Evaluation. AO1 Knowledge and understanding (max 3 marks) A diagram of a PPC: - the axes of the PPC are appropriately labelled, such as manufactured/agricultural goods or consumer/capital goods or Good A/Good B (1) - the PPC will need to touch both axes (1) - a position within a PPC is clearly shown. (1) AO2 Analysis (max 3 marks) An analysis of the significance of a position within/inside a PPC: - a position where an economy is using its resources inefficiently - where not all available resources are being utilised - where the output of both products is lower than it would be if all available resources were being used - there should be some reference to permanency. AO3 Evaluation (max 2 marks) Offers a valid judgement on whether such a position is likely to be permanent (1) to reach a conclusion. (1)	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
5	With the help of a production possibility curve (PPC) diagram(s), explain the difference between constant and increasing opportunity costs and consider how choices in deciding which type of goods to produce in the short run may influence future economic growth. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding, up to 3 marks for AO2 Analysis, and up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) - Accurately drawn and labelled production possibility diagram – curved to show (and be identified as) increasing opportunity cost (1) and – straight line to show (and be identified as) constant opportunity cost (1). N.B., axes must be labelled as products and not simply A and B or X and Y and the curves must touch both axes. - Explanation for the difference between constant and increasing opportunity cost. For example, constant opportunity cost demonstrates no change in the rate of trade off whereas increasing opportunity costs demonstrate an increase rate in the trade off between the two products (1). AO2 Analysis (max 3 marks) For example, future economic growth will be influenced by the choices an economy makes about the split between consumer spending and capital spending (1). If a country decides to focus its spending and output on consumer goods, then this may create growth in the short run but may not be sustainable (1). A decision to focus on spending and output of capital goods, this may have a short run negative impact on economic growth but should create more long-term sustainable growth as output may expand (1). AO3 Evaluation (max 2 marks) Up to 2 marks for valid evaluation which is clearly based on the analysis (1) leading to a valid conclusion (1).	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
6	With the help of a production possibility curve (PPC) diagram, explain the terms scarcity and choice and consider the extent to which every choice has an equal opportunity cost. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) An accurate PPC diagram – a concave or straight-line curve that meets both axes with appropriate and correctly labelled axes (1); accurate definition of opportunity cost (1) and how the scarcity in factor endowments/scarcity results and infinite wants results in choices having to be made (1). AO2 Analysis (max 3 marks) If the candidates uses a straight line PPC Uses the diagram to explain that scarcity is shown as a point on the straight line PPC (1); the opportunity cost of changing production combinations is shown as a movement along the PPC (1), the opportunity cost is equal due to the negative one gradient of the PPC (accept constant trade off although not strictly true) (1) If the candidates uses a concave PPC Uses the diagram to explain that scarcity is shown as a point on the straight line PPC (1); the opportunity cost of changing production combinations is shown as a movement along the PPC (1), the opportunity cost is not due to the increased trade-off the greater the proportion of resources allocated to the production of a good (1) AO3 Evaluation (max 2 marks) One mark for a valid explanation that the extent to which each choice has an equal opportunity cost depends on the shape of the PPC (1) or whether or not there are unused factors of production when the choice is being made (1). Please use a text box to show the mark split e.g., 3 3 1	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
7	An economy contains just two sectors, an agricultural sector and an industrial sector. With the help of a diagram, explain how a production possibility curve (PPC) can be used to show the concept of opportunity cost <u>and</u> consider whether opportunity cost is likely to remain constant if resources are reallocated from the agriculture sector to the industrial sector. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) An understanding of a PPC shown through an accurate diagram (1), correctly labelled axes with agriculture on one axis and industrial products on the other axis (1). Defines or demonstrates clear understanding of the concept of opportunity cost (1). AO2 Analysis (max 3 marks) Uses the PPC to explain that a movement along the curve involves a reallocation of resources (1), as resources are shifted between the two types of product, there is an opportunity cost involved (1), measured by a sacrifice of agricultural products as more industrial products are produced. (1) AO3 Evaluation (max 2 marks) Those resources least suited to agriculture will be redeployed first so there will be low opportunity costs (1) so opportunity costs will always rise as resources are increasingly reallocated from agriculture to industrial uses (1).	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

SECTION B

1.	C	11.	C	21.	B
2.	D	12.	D	22.	A
3.	B	13.	C	23.	A
4.	C	14.	D	24.	A
5.	D	15.	C	25.	C
6.	A	16.	C	26.	B
7.	C	17.	D	27.	A
8.	C	18.	D	28.	C
9.	D	19.	B	29.	C
10.	D	20.	C	30.	B